

Autonomous Theory Construction Laboratory

Lecture 7: Run Histories and Reproduction

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Overview

A hash-chained run history records every accepted and rejected proposal and supports deterministic reproduction.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$h_r = H(h_{r-1}, event_r)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$root_n = Merkle(h_1, \dots, h_n)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$artifact_r = Replay(env, data, candidate_r, seed_r)$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- RLP-12 complete
- Yun, ch. 7
- Replay Preservation Standard